An Economic Analysis of Financial Structure

Chapter 8

Econ 351 – Monetary Economics

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Preview

This chapter focuses on transaction costs and information asymmetry, including adverse selection and moral hazard.

- **Transaction costs** keep small investors out of securities markets; intermediaries reduce them through economies of scale and expertise.
- **Adverse selection** (before the transaction): the lemons problem.
- **Moral hazard** (after the transaction): the principal–agent problem and risk-taking by borrowers.

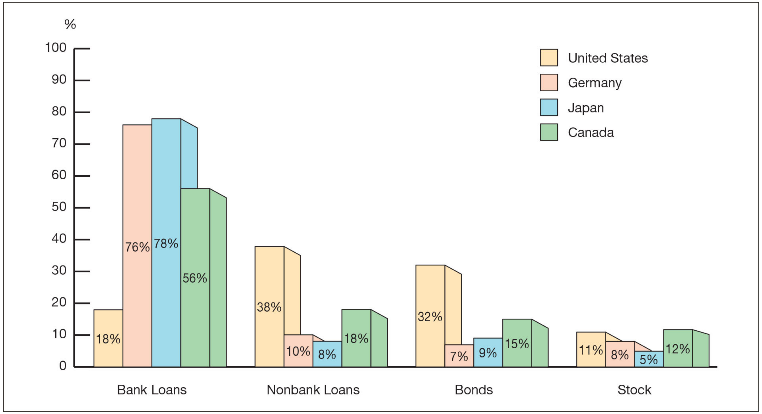
Outline

Basic Facts About Financial Structure

Transaction Costs

Information Asymmetry

Sources of External Funds for Businesses



Sources of external funds for nonfinancial businesses: the United States compared with Germany, Japan, and Canada.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Basic Facts About the Global Financial System

- **Stocks play only a limited role** in business financing.
- **Marketable securities** are less important than bank loans.
- **Indirect finance** (through financial intermediaries) dominates.
- **Banks** are the most important financial intermediaries worldwide.
- The financial system is **heavily regulated**.
- Only **large firms** can easily access securities markets.
- **Collateral** is essential for most lending.
- **Debt contracts** are subject to strict legal enforcement.

Outline

Basic Facts About Financial Structure

Transaction Costs

Information Asymmetry

What Are Transaction Costs?

Transaction Costs

Costs associated with **buying, selling, or managing financial assets**.

- Brokerage fees
- Legal and administrative costs
- Search and information costs

Why Do Transaction Costs Matter?

- They create a **barrier to entry** for small investors.
- High costs reduce **market efficiency**.
- Limited diversification increases **financial risk**.

For Small Investors

- Investment amounts are too small relative to fees.
- It is difficult to achieve diversification (portfolio theory).

How Intermediaries Reduce Transaction Costs: Economies of Scale

Two Ways Financial Intermediaries Reduce Transaction Costs

- **Economies of scale** $\rightarrow$ lower costs per transaction
- **Expertise in risk assessment** $\rightarrow$ smarter investment decisions

Economies of Scale

- Larger transactions $\Rightarrow$ lower cost per unit.
- Bigger investors face lower transaction costs.
- This explains the role of **mutual funds** and banks.

Examples

- Mutual funds pay lower fees per share.
- Banks spread fixed costs across many loans.

How Intermediaries Reduce Transaction Costs: Expertise

Expertise

Professional financial intermediaries reduce costs and risk through **expert knowledge and analysis**.

What Do Financial Experts Do?

- **Market research:** identify profitable investment opportunities
- **Credit evaluation:** assess borrower creditworthiness
- **Regulatory knowledge:** ensure compliance with laws

Example

- Banks analyze thousands of loan applicants.
- Individual investors lack the resources to do this.

Outline

Basic Facts About Financial Structure

Transaction Costs

Information Asymmetry

Information Asymmetry

Information asymmetry occurs when one party has more or better information than the other, affecting the outcome of a transaction.

Adverse Selection (Before the Transaction)

- High-risk borrowers are more likely to seek loans.
- Banks cannot rely only on interest rates when lending.

Moral Hazard (After the Transaction)

- Borrowers may use funds in unintended or riskier ways.
- Lack of monitoring increases financial risk.

Information Asymmetry in Insurance Markets

Adverse Selection

- Who is more likely to buy insurance?
  - Risk-averse individuals
  - Individuals with higher expected losses
- Insurance premiums may be priced too low if risk differences are hidden.

How Insurance Contracts Reduce the Problem

- **Information acquisition:** medical exams and risk screening
- **Incentive alignment:** deductibles and co-payments

Adverse Selection: The Lemons Problem

The Lemons Problem (Used-Car Market)

- Buyers cannot tell whether a used car is high quality (“peach”) or low quality (“lemon”).
- Sellers know the true quality $\Rightarrow$ **asymmetric information**.

How the Market Works

- Buyers offer an average price.
- Owners of good cars leave the market.
- Low-quality cars dominate the market.

The market shrinks and may fail because buyers lose confidence.

A used-car lot: which ones are lemons? Only the seller knows.

Wikimedia Commons, Bob Harvey, CC BY-SA 2.0

The Lemons Problem in Financial Markets

- Investors cannot distinguish between **good firms** and **bad firms**.
- Firms know their true quality $\Rightarrow$ **asymmetric information**.
- Investors offer an average price.

Market Impact

- Good firms exit because they are undervalued.
- Average quality declines $\Rightarrow$ risky firms dominate.
- Market participation falls, leading to possible market failure.

Information asymmetry helps explain why securities markets are not the primary source of business financing.

Solutions to the Lemons Problem

Tools that reduce **adverse selection**:

- **Private production of information:** credit rating agencies (e.g., Moody's, S&P) assess firm quality.
- **Government regulation:** disclosure rules improve market transparency.
- **Financial intermediation:** banks collect private information and screen borrowers.
- **Collateral and net worth:** firms with assets or high equity reduce investor risk.

These solutions reduce asymmetric information but do not eliminate it completely.

Standard & Poor's headquarters in New York: a private producer of information.

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Moral Hazard and the Principal–Agent Problem

The Principal–Agent Problem

- **Stockholders (principals)** own the firm but do not manage it.
- **Managers (agents)** control decisions and may not act in owners' best interests.
- Managers have weaker incentives to maximize profits.

Examples

- Excessive spending on perks or luxury offices.
- Managers pursuing personal goals (e.g., empire building).

Moral hazard in equity contracts: because ownership is separated from control, managers (agents) may act in their own interest rather than in the interest of stockholders (principals).

Reducing the Principal–Agent Problem

- **Monitoring (costly state verification)**
  - Stockholders can monitor managers, but monitoring is costly.
  - The **free-rider problem** limits individual incentives to monitor.
- **Government regulation**
  - Financial reporting rules improve transparency.
  - Fraud laws penalize misreporting and insider abuse.
- **Financial intermediation**
  - Venture capital and private equity actively monitor firms.

These methods reduce moral hazard but cannot eliminate it completely.

Debt vs. Equity: Why Debt Reduces Moral Hazard

Mechanism

- **Debt requires fixed payments.**
- Lenders monitor mainly when default occurs.
- Debt holders get paid first, reducing managers' ability to divert profits.

Why Debt Is Preferred

- Less monitoring needed
- Lower verification costs
- Lower moral hazard problems

Debt contracts reduce the frequency and cost of moral hazard, which helps explain why debt is more common than equity financing.

Moral Hazard in Debt Markets

- Borrowers may take **excessive risks** after receiving funds.
- Borrowers keep extra profits but face limited liability for losses.

Example

- A borrower promises a safe investment but chooses a risky project instead.
- If the gamble succeeds, the borrower keeps the gains.
- If it fails, the lender bears most of the losses.

Lenders become more cautious, reducing loan supply and capital availability.

Reducing Moral Hazard in Debt Contracts

Net Worth and Collateral

- Borrowers with high net worth or pledged collateral have more to lose.
- This creates **“skin in the game”** and reduces risk-taking.

Monitoring and Restrictive Covenants

- Lenders impose legal restrictions on how funds are used.
- Common covenants:
  - Discourage risky behavior
  - Encourage desirable actions
  - Protect collateral value
  - Require financial transparency

These tools reduce risk-taking incentives but require ongoing monitoring.

Why Banks Are Crucial in Debt Markets

Why Banks Matter

- Banks reduce the **free-rider problem** by making private loans.
- Banks **directly monitor borrowers** and enforce loan conditions.
- Banks manage risk more effectively than individual investors.

Banks dominate debt markets because they actively monitor borrowers and enforce loan contracts. Because their loans are private and not traded, other investors cannot free-ride on the bank's monitoring.

Chapter summary

- Indirect finance through banks dominates business financing; stocks and marketable securities play a limited role, and only large firms access securities markets easily.
- Transaction costs shut small investors out; intermediaries reduce them through economies of scale and expertise.
- Adverse selection (the lemons problem) is reduced by private information production, regulation, intermediation, and collateral and net worth.
- Moral hazard in equity (principal–agent problem) explains why debt is preferred; moral hazard in debt is reduced by net worth, collateral, covenants, and monitoring.
- Banks are crucial because private loans avoid the free-rider problem in monitoring.